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A Data-Driven Lending Strategy through Credit Risk Analytics

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RevoU FSDA batch OCT25

DEEPP

INDUSTRY

Finance & Banking

DATASET

Lending Club Loan Dataset

https://www.kaggle.com/datasets/musabhawai/lending-club-loan-dataset?select=lending_club_loan_two.csv

WHY?

- Growth & Job Opportunities
- Business Impact
- Dataset Availability

WHY?

- Relevance to Industry
- Rich & Structured Data
- Business Problem Alignment
- Portfolio Value
- Availability & Accessibility

OUR COMPANY

WHO WE ARE?

LendingClub is a P2P lending platform turned into digital bank in USA.

WHAT WE DO?

The company wants to reverse declining profits by reducing defaults

HOW WE DO?

Data analysts play a critical role in building propensity models for better credit risk assesment.

BUSINESS MODEL

Intermediation

Intermediaries between savers & borrowers

Revenue Model

- Interest spreads
- Fees
- Investment returns

Regulation

Controlled by regulators (FDIC, OCC, Federal Reserve)

Digital Transformation

- Mobile banking
- Automation & AI

Expanded Services

- Debit cards
- Digital payments.
- Banking-as-a-Service (BaaS)

KEY BUSINESS PROCESSES

Customer Onboarding & KYC

Loan Application

Credit Assessment

Fraud Detection & Risk Management

Loan Approval & Disbursement

Continuous Improvement

Customer Engagement & Retention

Financial Reporting & Profitability Analysis

Revenue Generation

Repayment Monitoring & Collections

CUSTOMER INTERACTION POINTS

Digital Platforms

- Website
- Mobile App

Customer Support

- Live chat
- Email
- Call center

Transaction

- Loan repayment portals.
- Transfers between accounts or to external banks.
- Debit card
- Digital payment

Marketing

- Email campaigns
- Push notifications
- Personalized recommendations within the app.
- Financial education content

Regulatory

- KYC (Know Your Customer)
- Identity checks

DATA-DRIVEN DECISION MAKING IN DIGITAL BANKING & LENDING

CREDIT RISK ASSESSMENT

Machine learning models to evaluate loan applications.

FRAUD DETECTION & COMPLIANCE

Anomaly detection models to flag suspicious activity..

CUSTOMER RETENTION & ENGAGEMENT

Repayment behavior, churn risk, and customer satisfaction

CUSTOMER SEGMENTATION, ACQUISITION & MARKETING

Analyze customer to target the right segments.

CAMPAIGN, PRODUCT & PRICING OPTIMIZATION

New financial products.

OPERATIONAL EFFICIENCY

Dashboards and KPIs

PROJECT SCOPE

This project analyzes credit risk for LendingClub, a P2P lending platform turned into digital bank in USA., using a comprehensive loan dataset (2007-2016) containing 396,030 entries across 27 variables.

It focuses on factors influencing loan defaults, such as loan amount, interest rate, and annual income to build Prospensity Model for better credit risk assesment.

However, the analysis will not cover marketing strategies or operational aspects outside of credit risk.

STAKEHOLDERS

D
Decision-maker

Chief Lending Officer

A
Accountable

Credit Risk Analytics Director

R
Responsible

Credit Risk and Pricing Analyst

C
Consulted

Credit Specialist

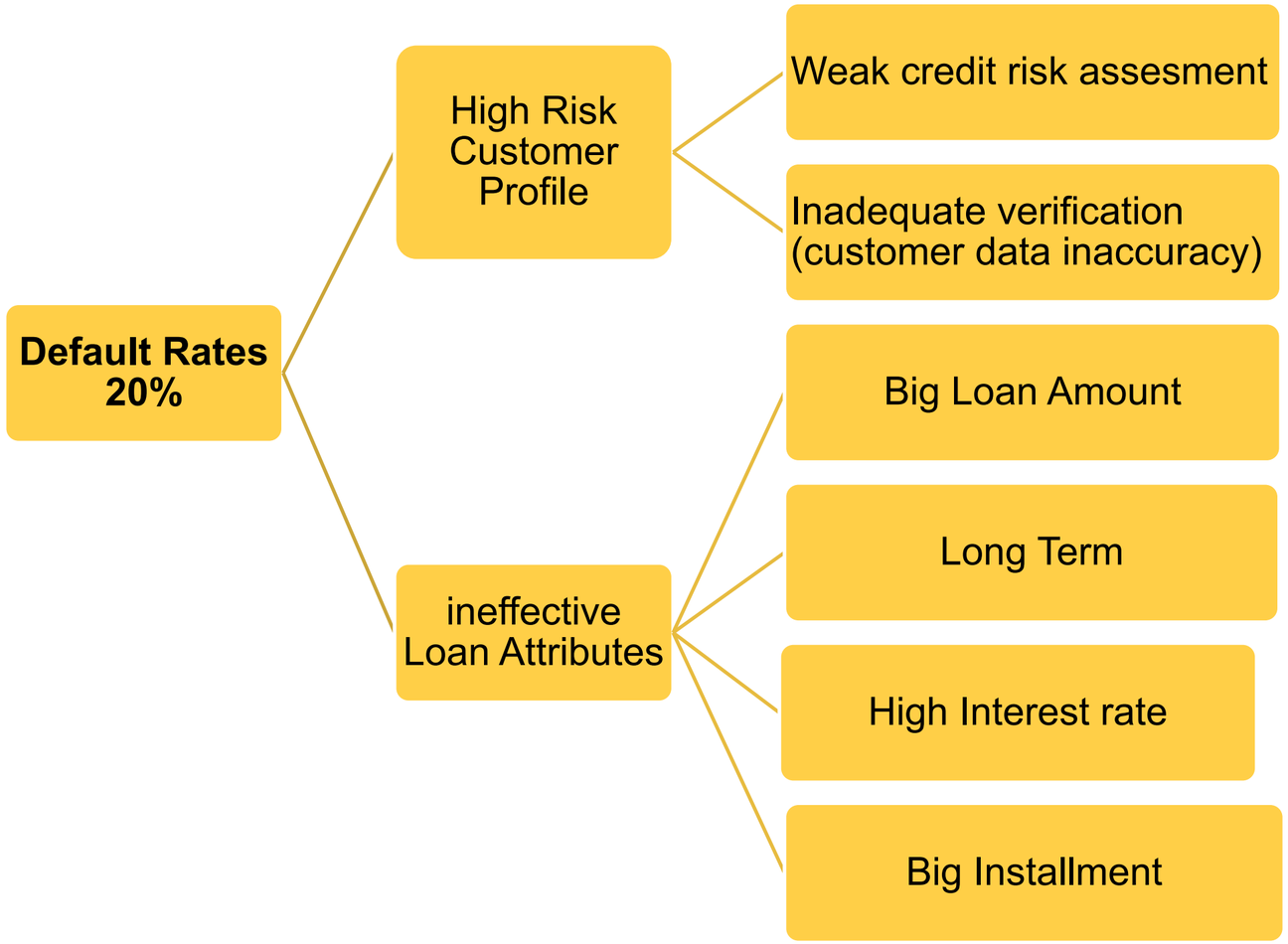
I
Informed

1. Top Management
2. Compliance, Policy & Regulatory Affairs

PROBLEM STATEMENT

We aim to identify the key factors influencing default rates in our P2P lending platform. Our goal is to reduce default rates from 20% to below 5% over the next 12 months by developing a better credit risk assessment.

S	Specific – target a specific area for improvement	Default rates
M	Measurable – quantify or show an indicator of progress Applied to the ideal state:	Reducing default rates from 20% to below 5%
A	Achievable – they need to be agreed, to be attainable and able to be implemented	Dataset of 396k records is large enough for prospensity model
R	Realistic – states what results can realistically be achieved, given available resources	Targets are aligned with industry benchmarks and organizational capacity, balancing ambition with feasibility
T	Time-bound - there need to be deadlines, but are they reasonable?	12 months



PRIORITY	ROOT CAUSE	HYPOTHESIS	METRICS
1	Weak credit risk assesment	If scoring & risk assessment are improved, then default rates will decrease.	Default rate
2	Inadequate verification (customer data inaccuracy)	If verification processes are enhanced, then default rates will decrease.	Default rate
3	High Interest rate	If average interest rate is decreased, then probability of customers failing to pay will be decrease.	Default rate Interest rate
4	Big Loan Amount	If average loan amount is decreased, then probability of customers failing to pay will be decrease.	Default rate Loan Amount
5	Big Installment	If average installment is decreased, then probability of customers failing to pay will be decrease.	Default rate Installment
6	Long Term	If term is longer, then probability of customers experiencing finansial downturn will affect customer's capacity	Default rate Term

ROOT CAUSES ANALYSIS & HYPOTHESIS

KPI

KPI	TARGET (in 12 months)
Default rate	< 5%

METHODOLOGY

DATA COLLECTION

- CUSTOMER PROFILE
- LOAN ATTRIBUTE

DATA CLEANING

HANDLING :

- MISSING VALUES
- DUPLICATES
- OUTLIER

EXPLORATORY DATA ANALYSIS

It can help answer questions like, “Who are our customers?” and “What are their characteristics?”

PROSPENSITY MODEL : LOGISTIC REGRESSION

Analyze historical data to identify factors that significantly influence default rate

DATASET STRUCTURE OVERVIEW

Number of Entries (Rows):

- Total entries: 396,030

Number of Columns:

- Total columns: 27

NO.	COLUMN NAME	DESCRIPTION	NON-NULL COUNT	DATA TYPE
0	loan_amnt	Amount of loan requested	396,030	float64
1	term	Term of the loan (e.g., 36 months, 60 months)	396,030	object
2	int_rate	Interest rate of the loan	396,030	float64
3	installment	Monthly payment for the loan	396,030	float64
4	grade	Loan grade (A, B, C, etc.)	396,030	object
5	sub_grade	Sub-grade of the loan (e.g., A1, A2)	396,030	object
6	emp_title	Employment title of the borrower	373,103	object
7	emp_length	Length of employment in years	377,729	object
8	home_ownership	Home ownership status (e.g., OWN, MORTGAGE, RENT)	396,030	object
9	annual_inc	Annual income of the borrower	396,030	float64
10	verification_status	Verification of income (Verified, Not Verified)	396,030	object
11	issue_d	Date the loan was issued	396,030	object
12	loan_status	Current status of the loan (Fully Paid, Charged Off)	396,030	object
13	purpose	Purpose of the loan (e.g., Debt Consolidation, Home Improvement)	396,030	object
14	title	Loan title	394,274	object
15	dti	Debt-to-Income ratio	396,030	float64
16	earliest_cr_line	Date of the borrower's earliest credit line	396,030	object
17	open_acc	Number of open credit accounts	396,030	float64
18	pub_rec	Number of derogatory public records	396,030	float64
19	revol_bal	Total credit revolving balance	396,030	float64
20	revol_util	Revolving credit utilization	395,754	float64
21	total_acc	Total number of credit accounts	396,030	float64
22	initial_list_status	Initial listing status (W, F for whole loans)	396,030	object
23	application_type	Type of application (individual or joint)	396,030	object
24	mort_acc	Number of mortgage accounts	358,235	float64
25	pub_rec_bankruptcies	Number of public record bankruptcies	395,495	float64
26	address	Address of the borrower	396,030	object

Full Analysis & Python Code available via Google Colab Link :
https://colab.research.google.com/drive/17iva4NAA-EU3cG0bJcW1jOwWR8HsJ-Y2#scrollTo=7r9DyUG8iKeS

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DATA CLEANING

Data cleaning	Column title	Before	After	Rationale
Missing values	emp_title	22927 rows	Keep missing values	Irrelevant data
	emp_length	18301 rows	Impute with 0	No formal jobs
	title	1756 rows	Keep missing values	Irrelevant data
	revol_util	276 rows	Impute with 0	The customer has not used his/her credit
	mort_acc	37795 rows	Impute with 0	The customer has no mortgage
	pub_rec_bankruptcies	535 rows	Impute with 0	The customer has no history of bankrupties
Duplicates	-	0 rows	-	-
Data Types	Term	String	Integer term (months)	Column should be numerical value to be included in building prospensity model
	emp_length	String	Integer emp_length (years)	Column should be numerical value to be included in building prospensity model
	issue_d	String	Datetime	Column should be datetime
	earliest_cr_line	String	Datetime	Column should be datetime

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DATA CLEANING

Data cleaning	Column title	Before	After	Rationale
Outlier	loan_amnt	Outlier detected Total data entry = 396.030	Add outlier flag column. Total data entry without outlier = 350.915 Outlier removed = 45.115 rows	Outliers can hinder prospensity model & EDA
	int_rate			
	Installment			
	annual_inc			
	DTI			
	open_acc			
	pub_rec			
	revol_bal			
	revol_util			
	total_acc			
	mort_acc			
	pub_rec_bankruptcies			
	term (months)	Only 2 unique values in this column : 36 & 60 months. 60 months is detected as outlier	Keep outlier	Used in analysis of default rate comparison between kredit term
	emp_length (years)	No oulier	-	-

EXPLORATORY DATA ANALYSIS

LOAN PURPOSE

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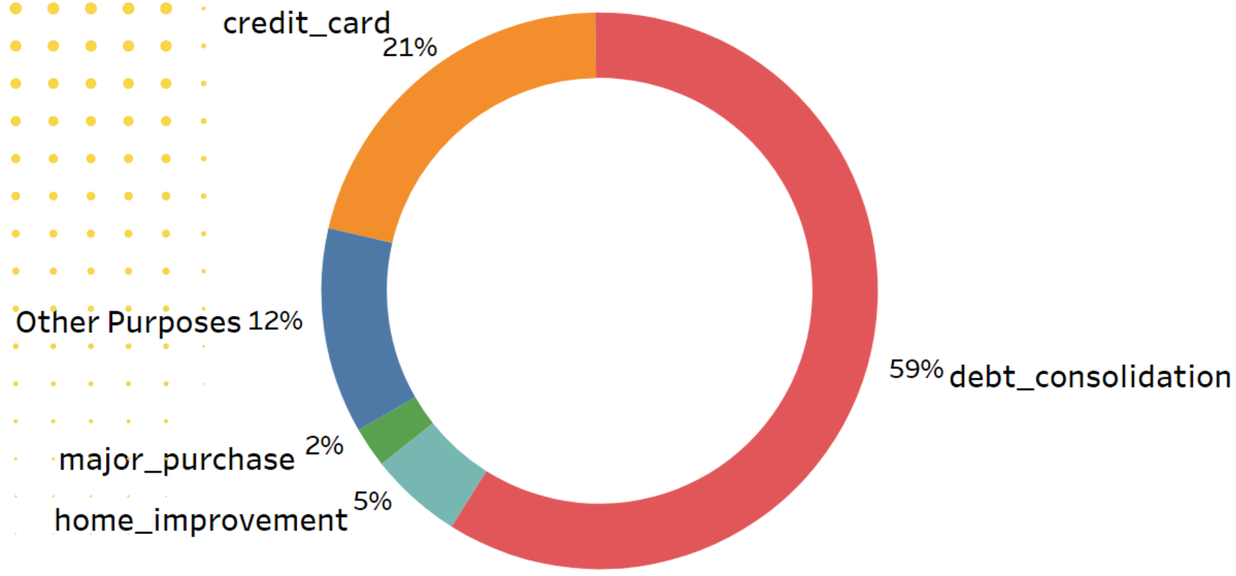
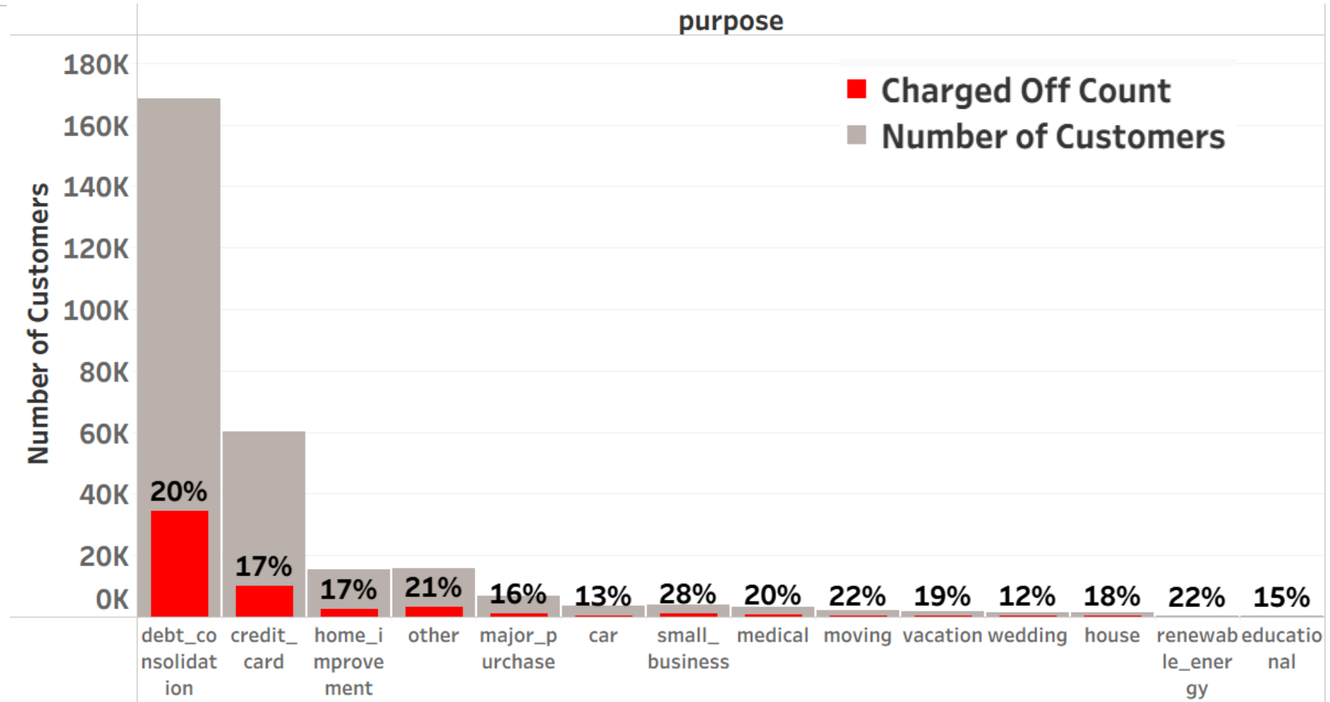
TOP LOAN PURPOSES

TOP 5 LOAN PURPOSES



LOAN STATUS BY PURPOSE



Top lending purpose is debt consolidation (> 50%) with high default rate.

The company may consider limit loan with this purpose & exploring other low risk purpose while fixing high default rate problem

Fully paid percentage is higher than charged off in all purpose.

However, charged off percentage is high (>5%) in all purpose.

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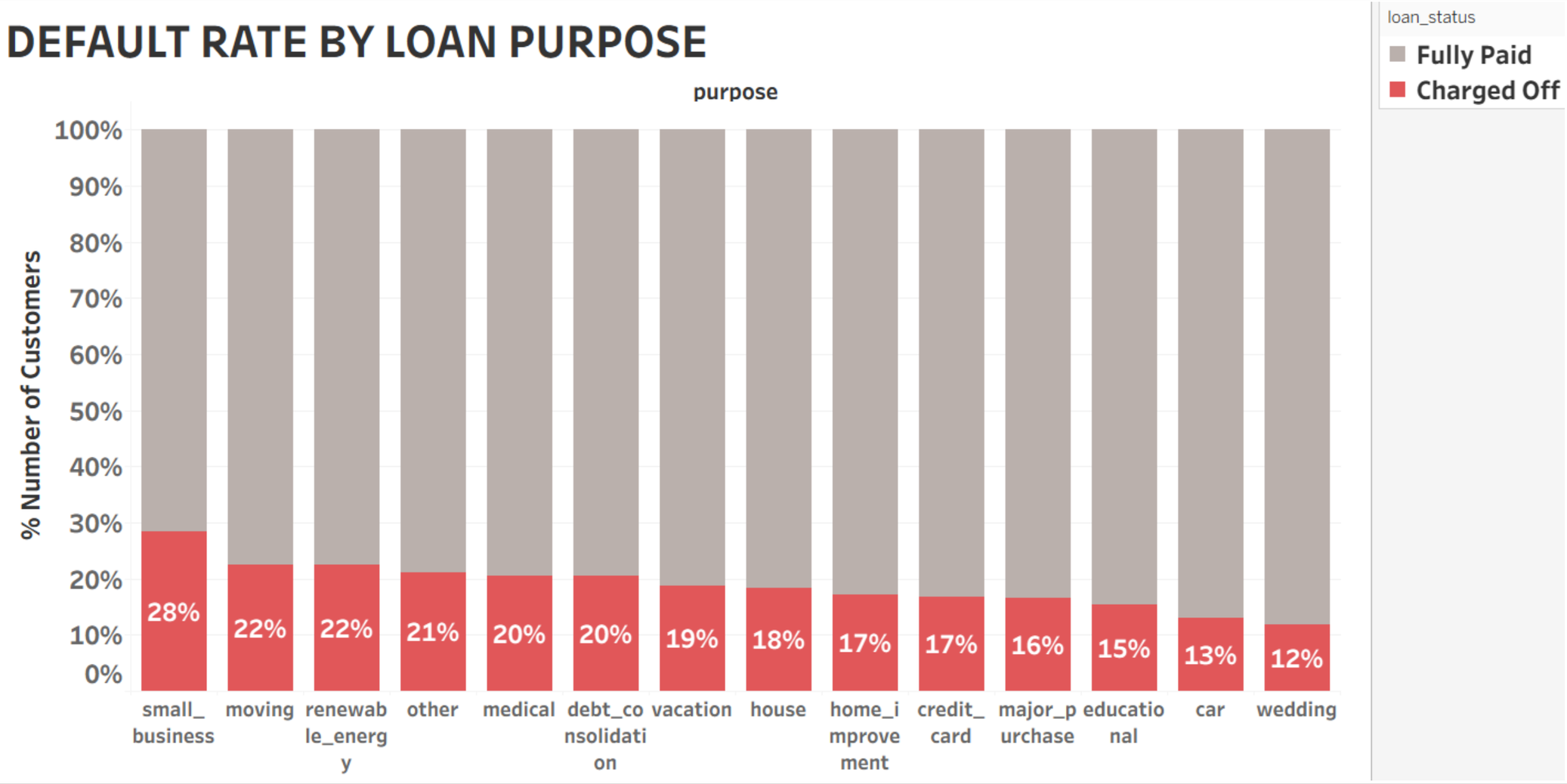
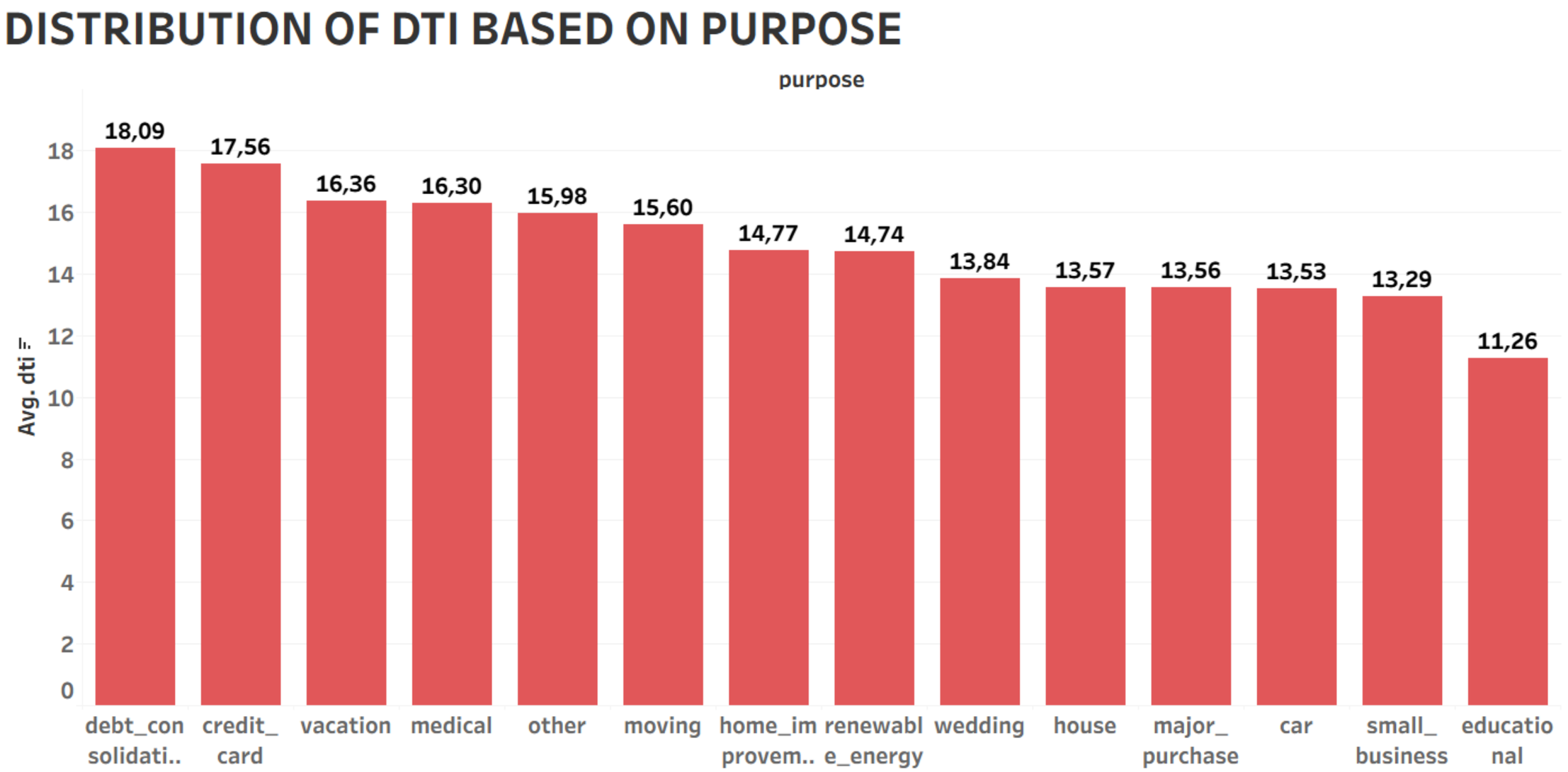
LOAN PURPOSES VS DEFAULT RATES



The company may consider limit loan purpose with higher charged off percentage like small business & moving.

And exploring other lower risk purpose like wedding, car, education.

DISTRIBUTION OF DTI BASED ON PURPOSE



The company may consider limit loan purpose with higher DTI average like debt consolidation, credit card, vacation. And exploring other lower risk purpose like car, education.

EXPLORATORY DATA ANALYSIS

TERM

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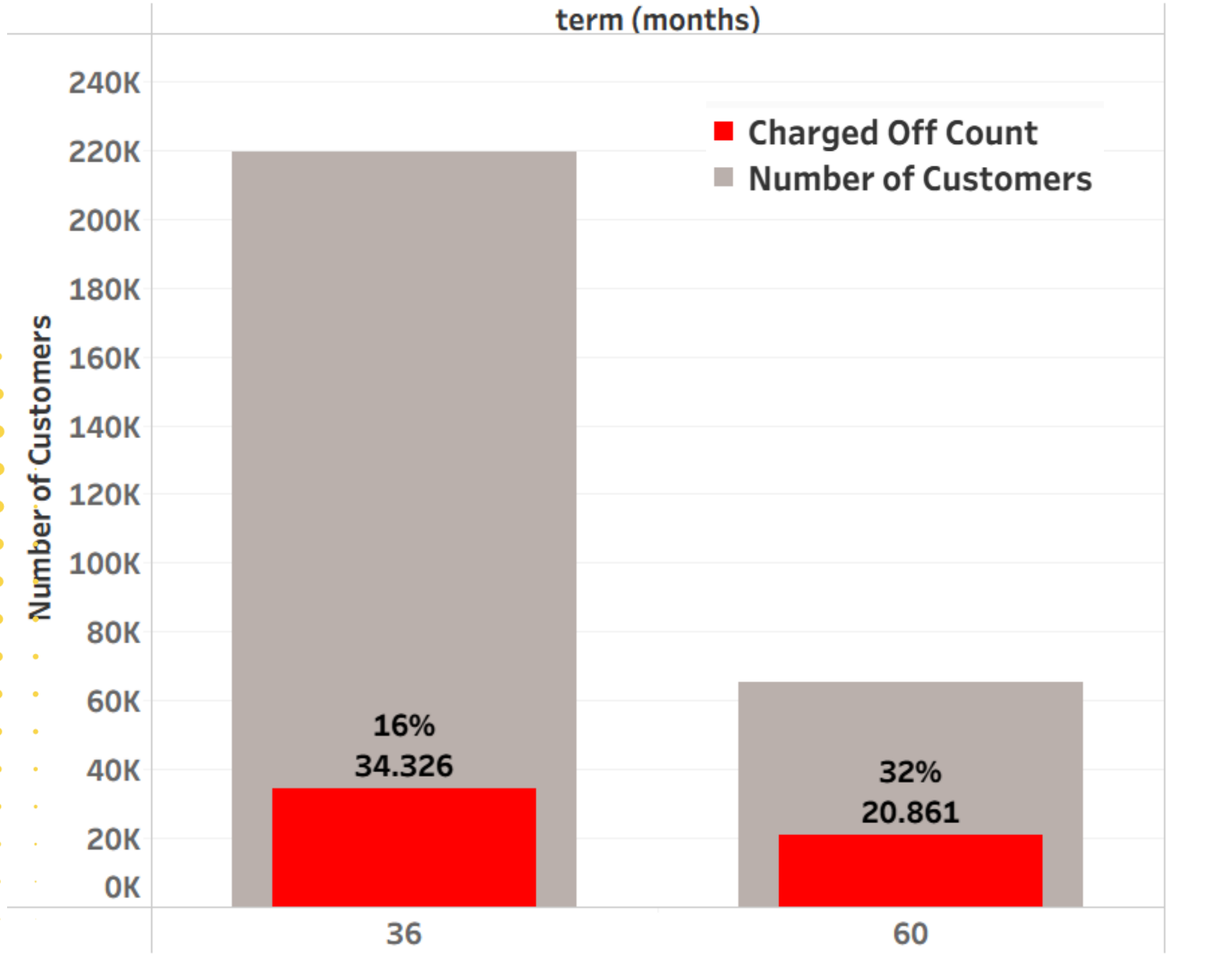
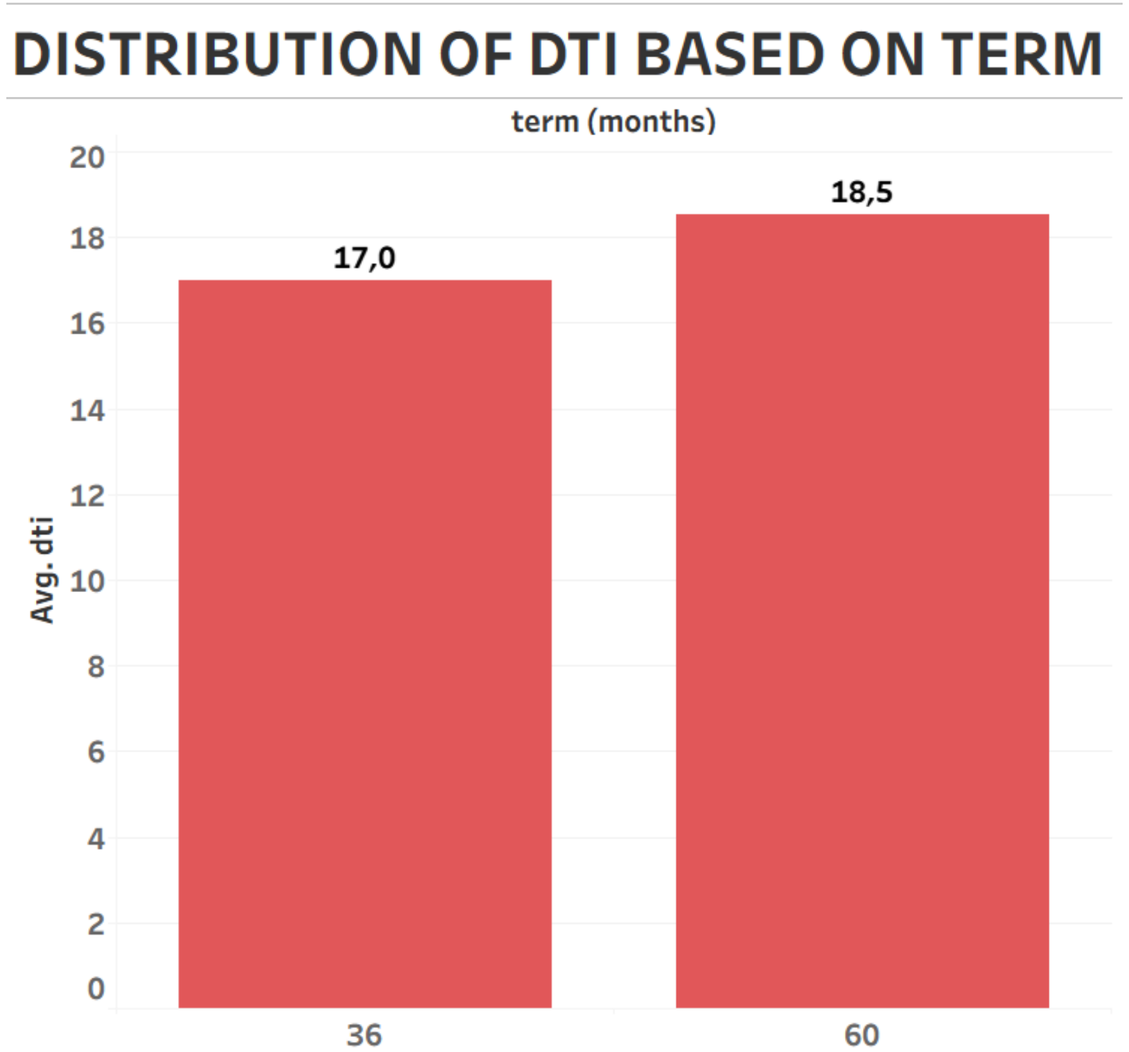
DISTRIBUTION OF LOAN STATUS BY TERM





Charged off percentage & DTI Average is higher in 60 months term than 36 months

Longer term means borrower at more risk having finansial change (e.g. Losing job) during term

EXPLORATORY DATA ANALYSIS

GRADE

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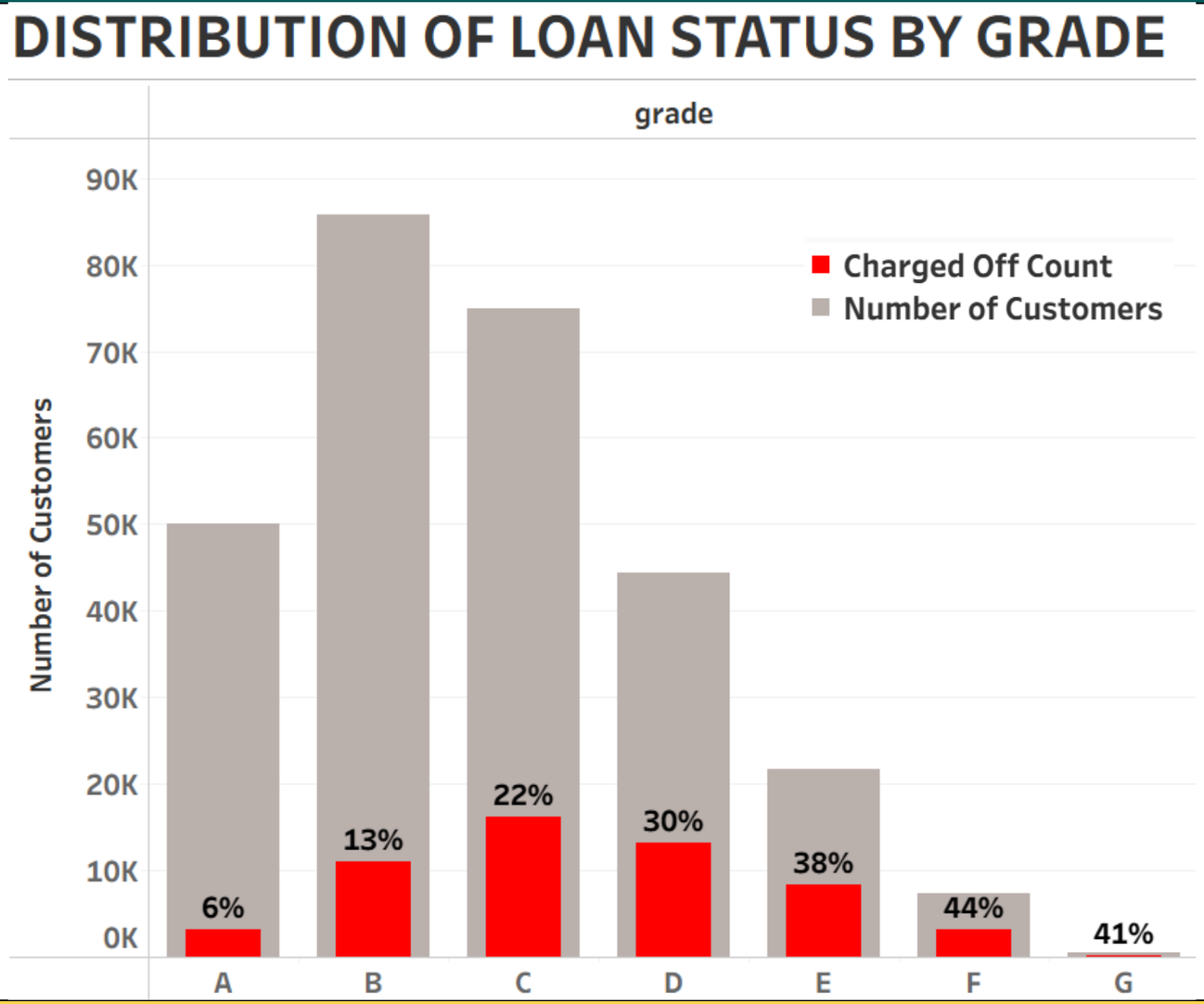
DISTRIBUTION OF LOAN STATUS BY GRADE



Higher grade (A,B,C) show better default rate than lower grade (D,E,F,G).

Higher grade (A,B,C) already show more loan approved than lower grade (D,E,F,G) with highest loan issued for grade B & C.

However, higher grade (A,B,C) still show relatively high default rate (>5%)

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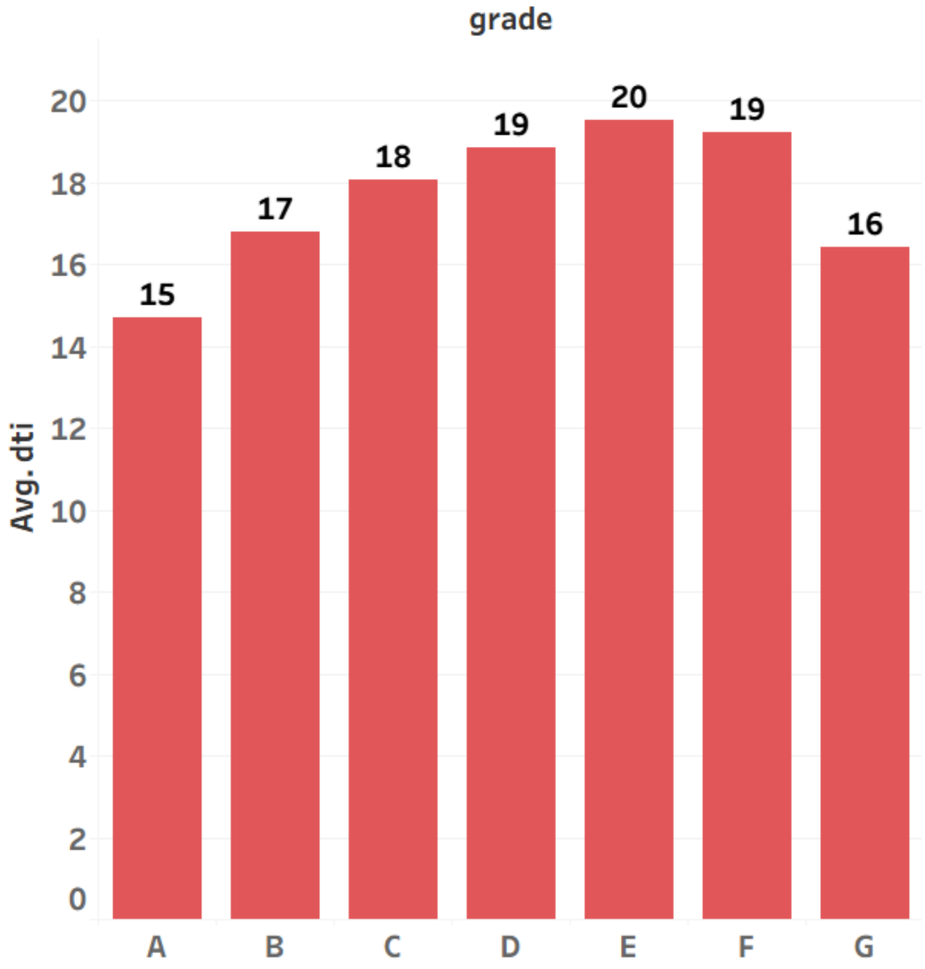
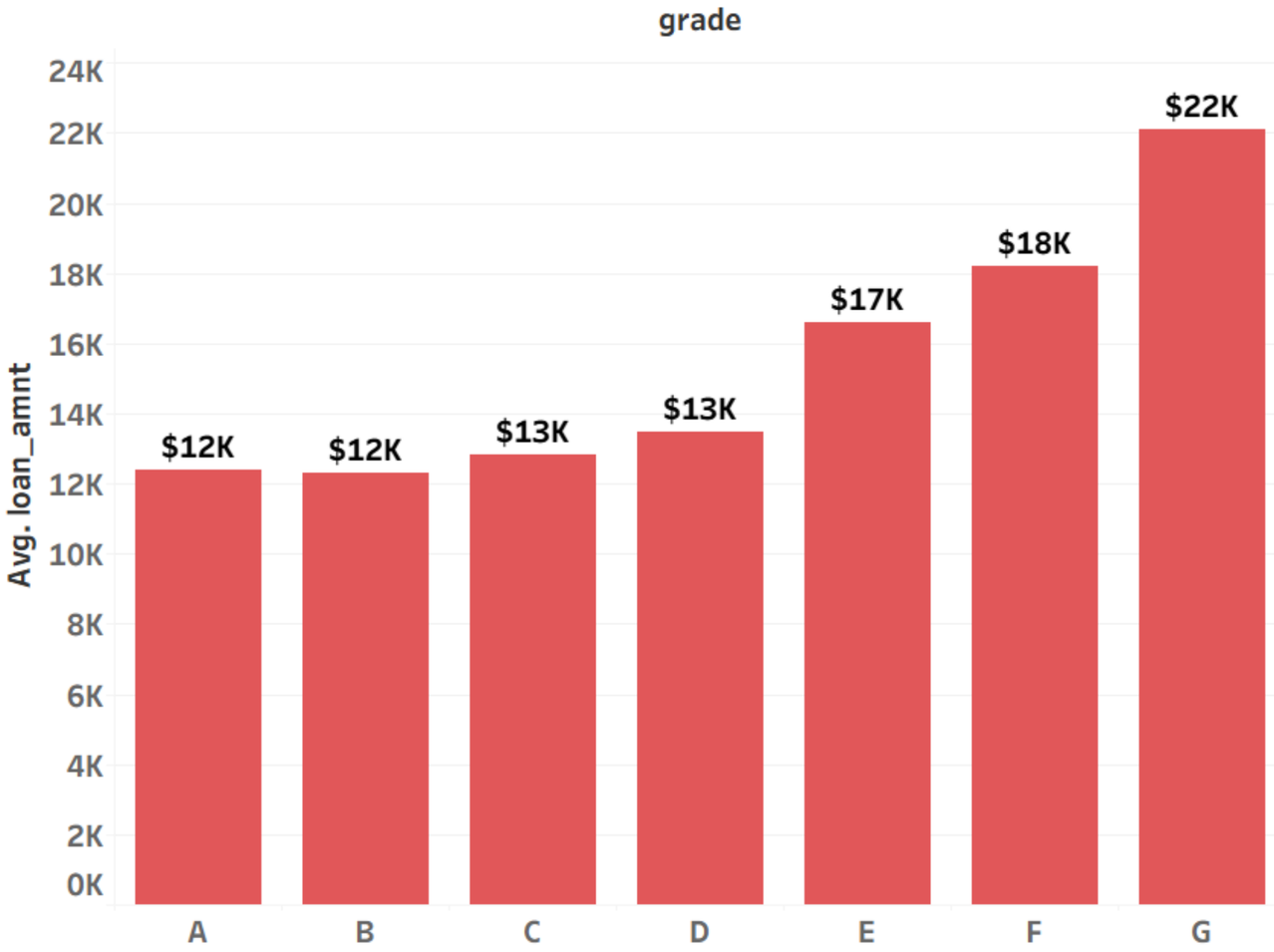
GRADE COMPARISON

DISTRIBUTION OF DTI BY GRADE



DISTRIBUTION OF LOAN AMOUNT BY GRADE



Higher grade (A,B,C) show lower DTI & Loan Amount

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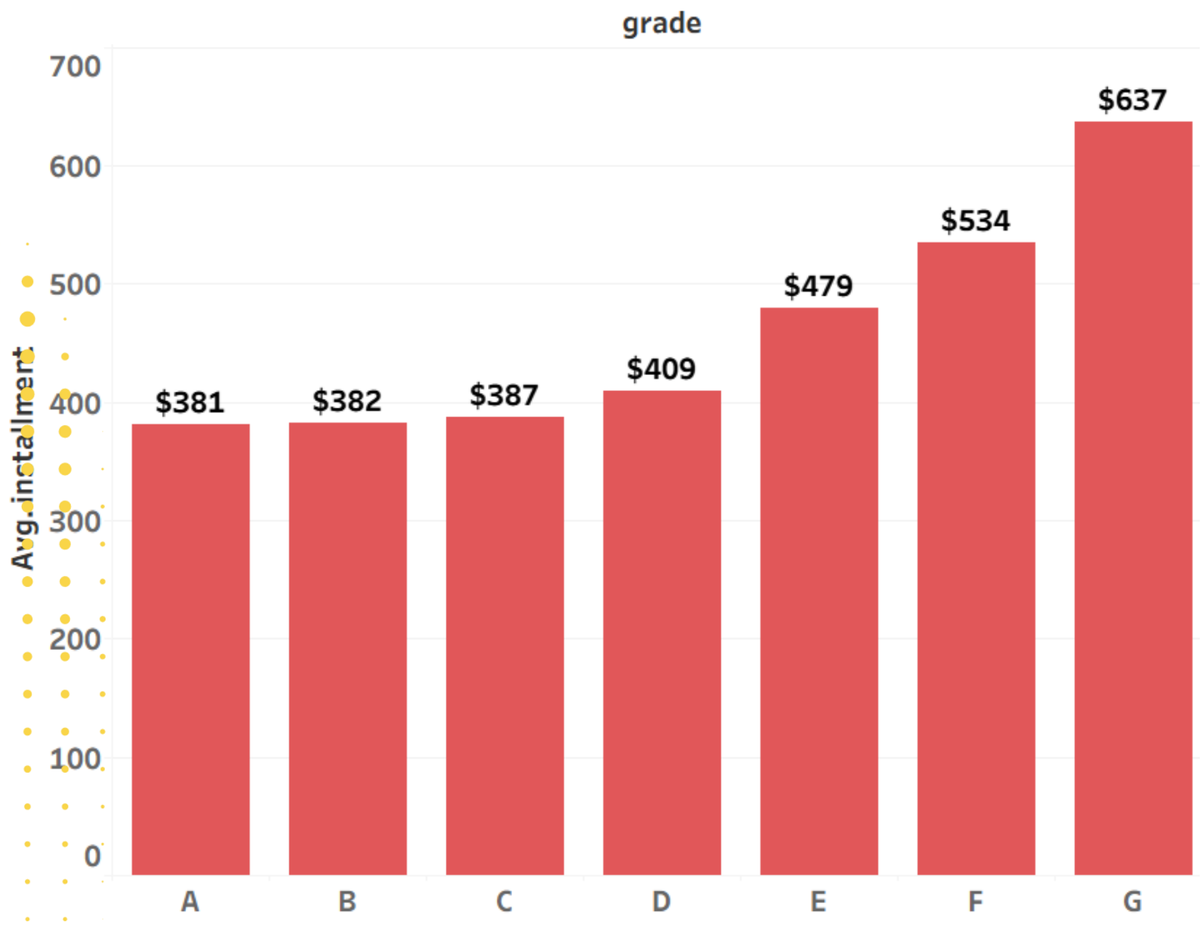
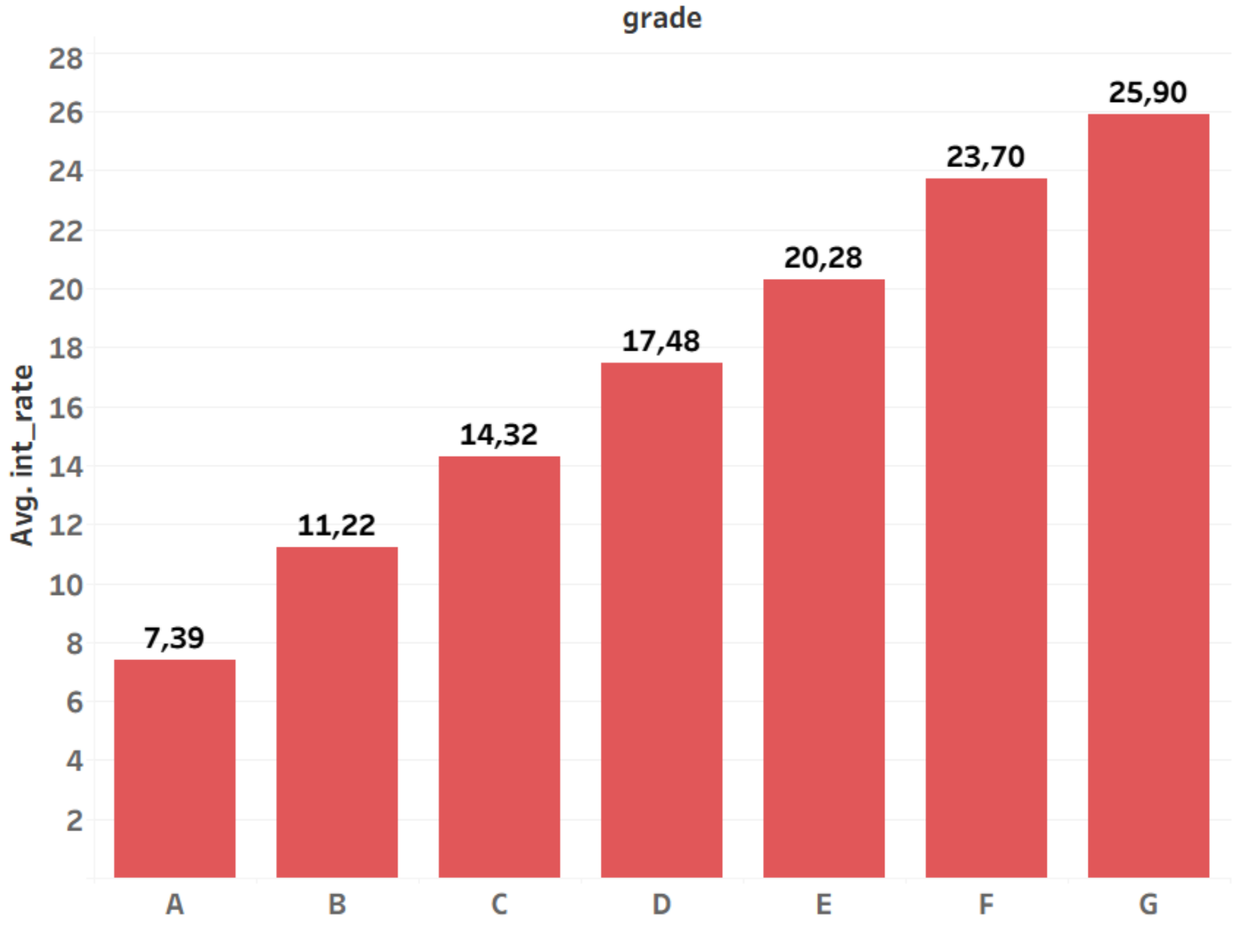
GRADE COMPARISON

DISTRIBUTION OF INSTALLMENT BY GRADE



DISTRIBUTION OF INTEREST RATE BY GRADE

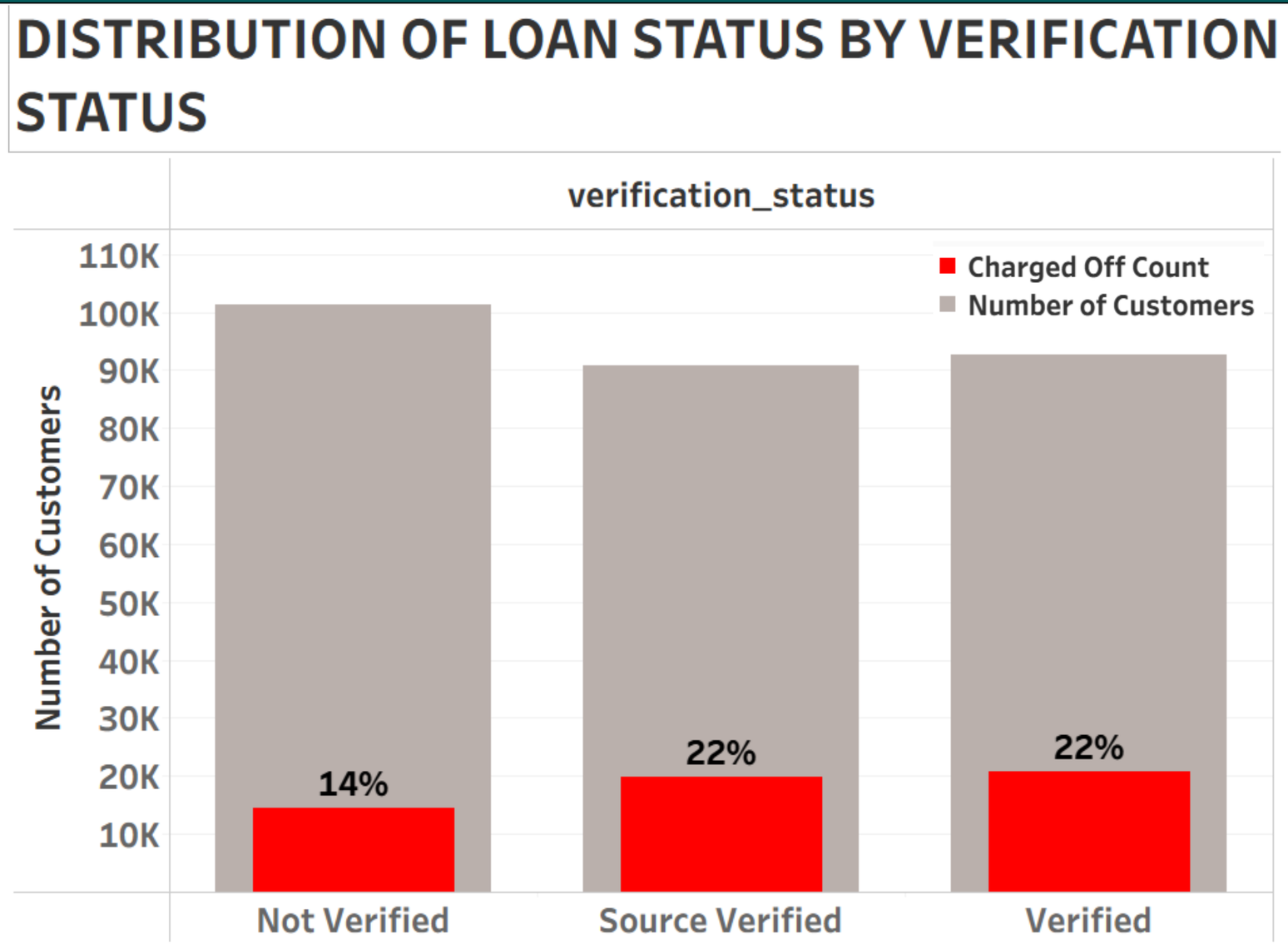


Higher grade (A,B,C) show lower Installment & Interest Rate

EXPLORATORY DATA ANALYSIS

VERIFICATION STATUS

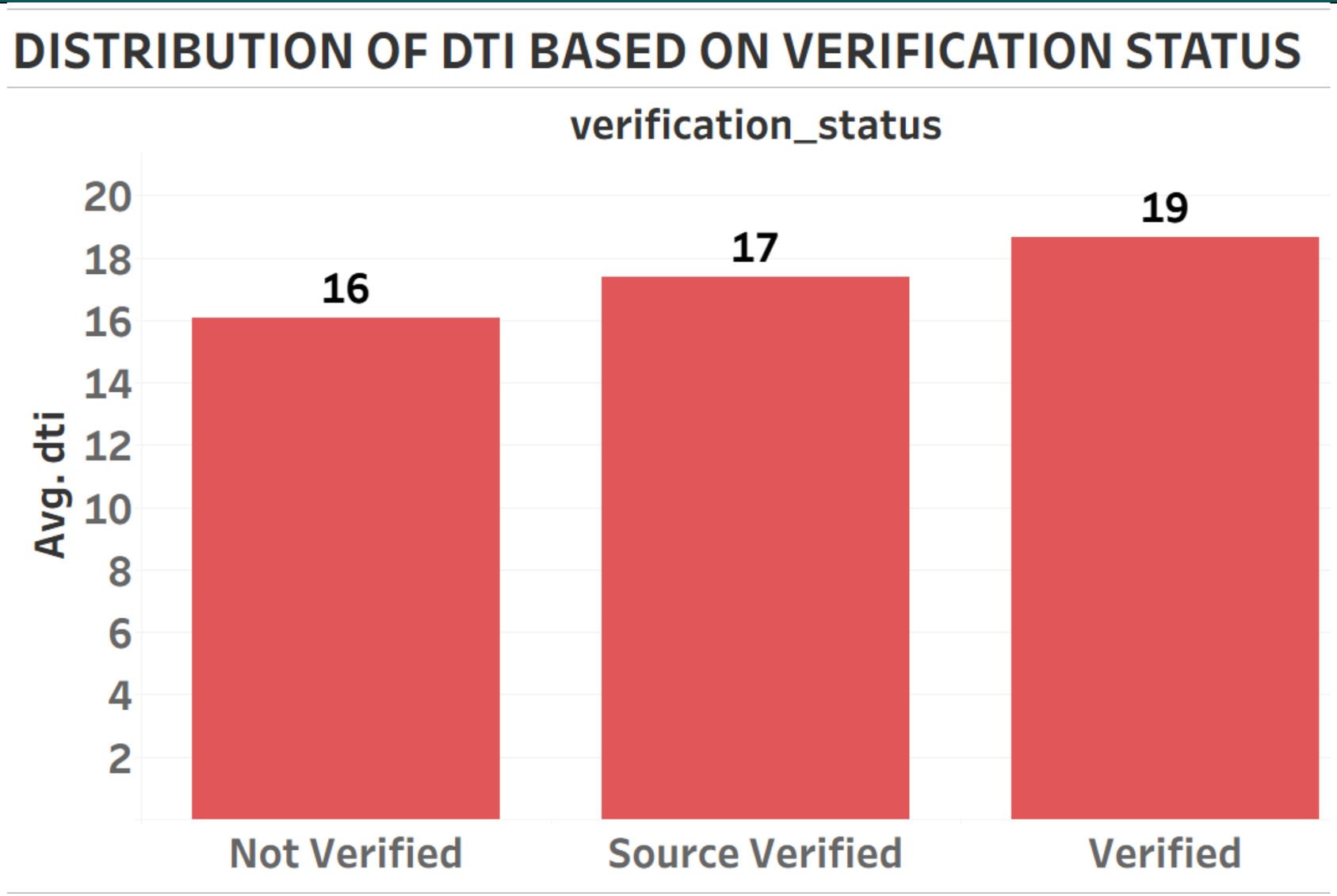
DISTRIBUTION OF LOAN STATUS BY VERIFICATION STATUS



Default rate in not verified is lower than verified.

However, overall default rate are high (>5%) in all verification status.

DISTRIBUTION OF DTI BASED ON VERIFICATION STATUS



Average DTI is higher on verified than not verified

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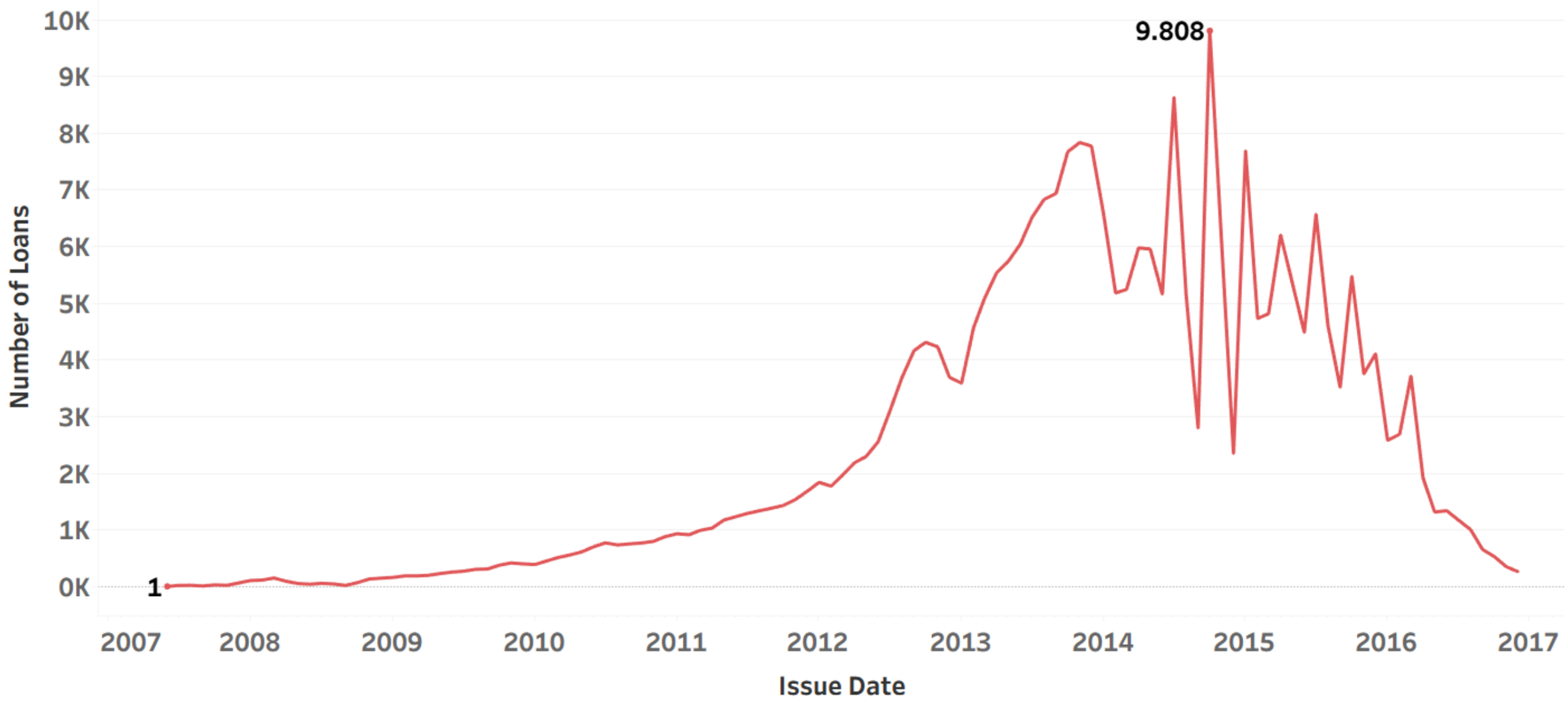
NUMBER OF LOANS ISSUED PER MONTH

NUMBER OF LOANS ISSUED PER MONTH



Lending Club CEO abruptly resigns amid $22m loan investigation

- Renaud Laplanche quit Monday as company's stock plummeted 25%
- Loan issued by company violated investor's instructions, statement said

Renaud Laplanche resigned as CEO of online loan company, Lending Club, on Monday. Photograph: Slaven Vlasic/(Credit too long, see caption)

Significant grow in 2012-2014 reflects the "Fintech Golden Age" where rapid expansion & massive investment toward fintech company was happening.

Significant drop in 2014-2017 probably because of internal factor (e.g. strict loan policy while company trying to address rising defaults problem) or external factor (e.g. Lending Club Scandal in 2015-2016)

PROSPENSITY MODEL :
LOGISTIC REGRESSION

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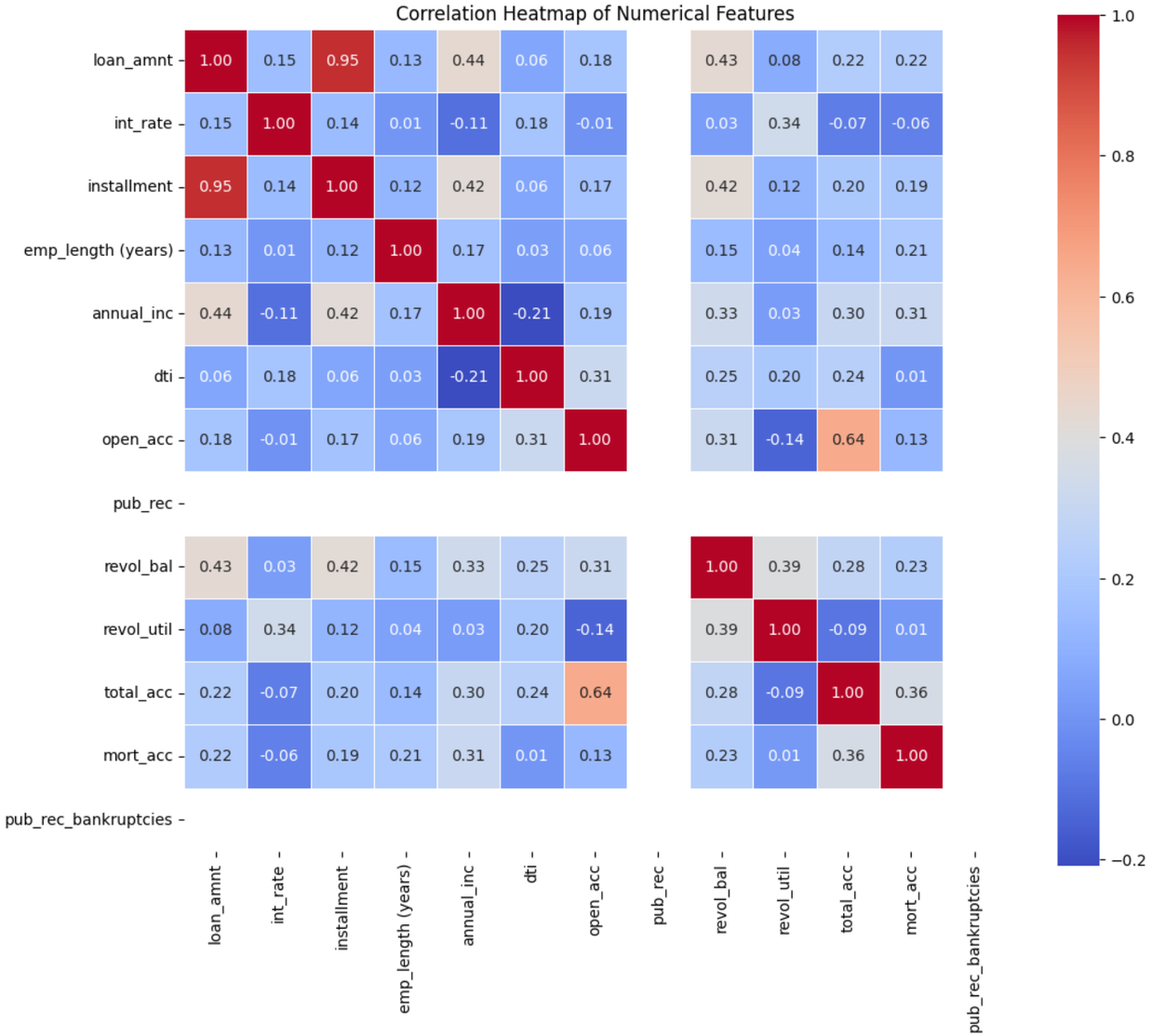
CORRELATION ANALYSIS



Loan Amount has high positive correlation with Installment

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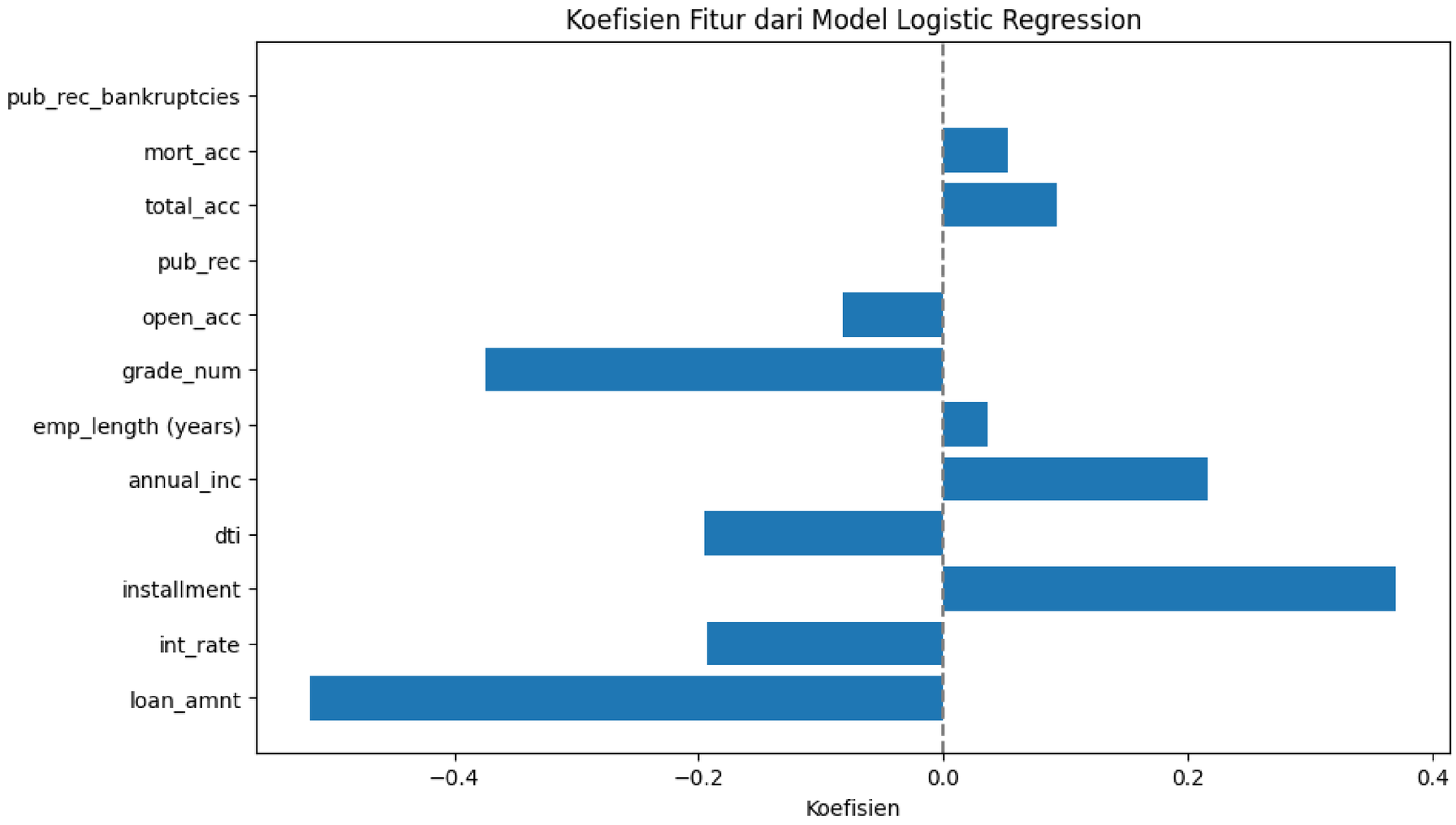
FACTORS INFLUENCING LOAN STATUS



Loan amount, interest rate, DTI, grade, open account impact negatively to default rate (higher default rate)

Installment, annual income, employment length, total account, mortgage account impact positively to default rate (lower default rate)

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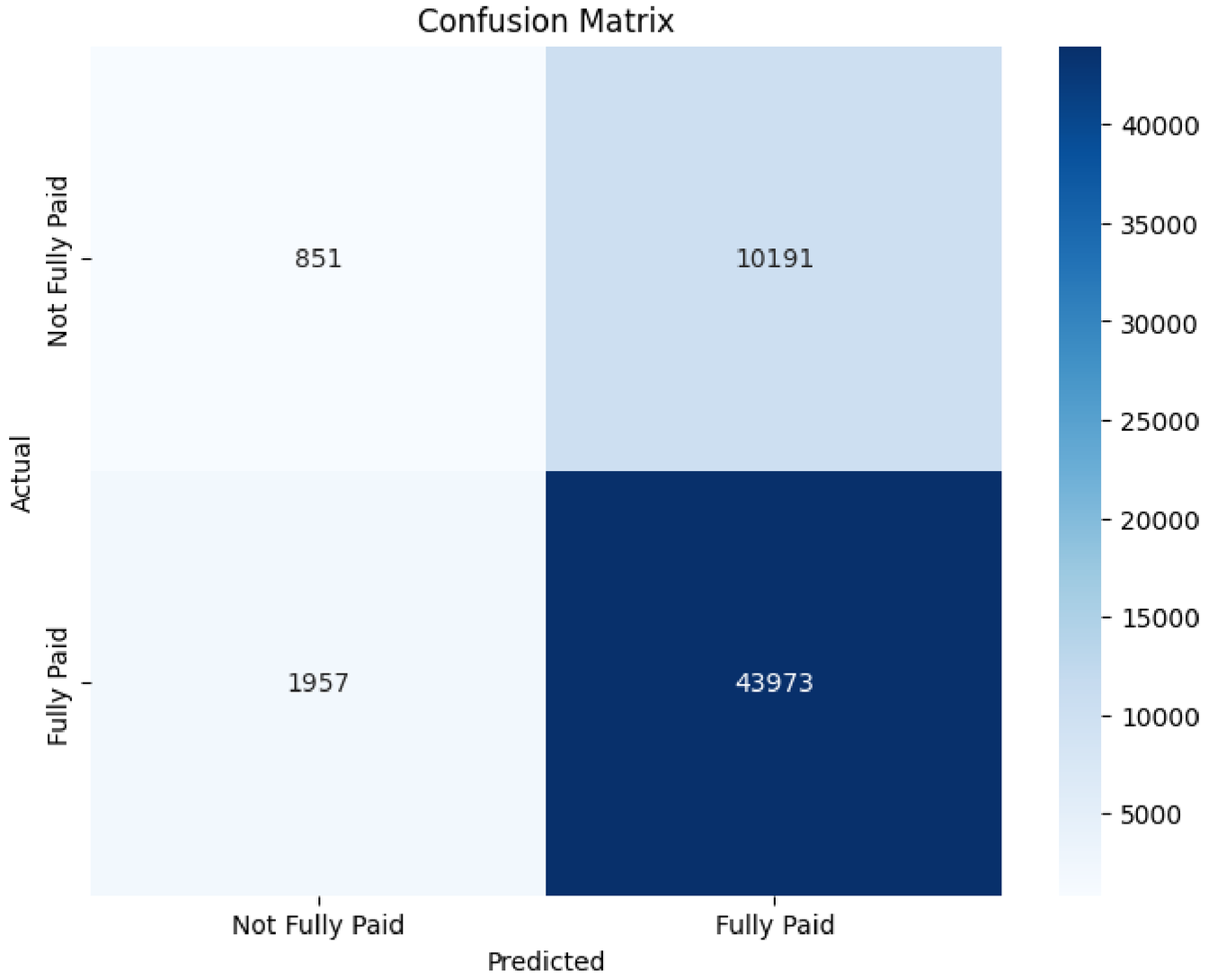
MODEL EVALUATION



01

ACCURACY = 84%

Reliable model

02

PRECISION = 67%

Need improvement in false positives

03

RECALL = 80%

Good job in true positives

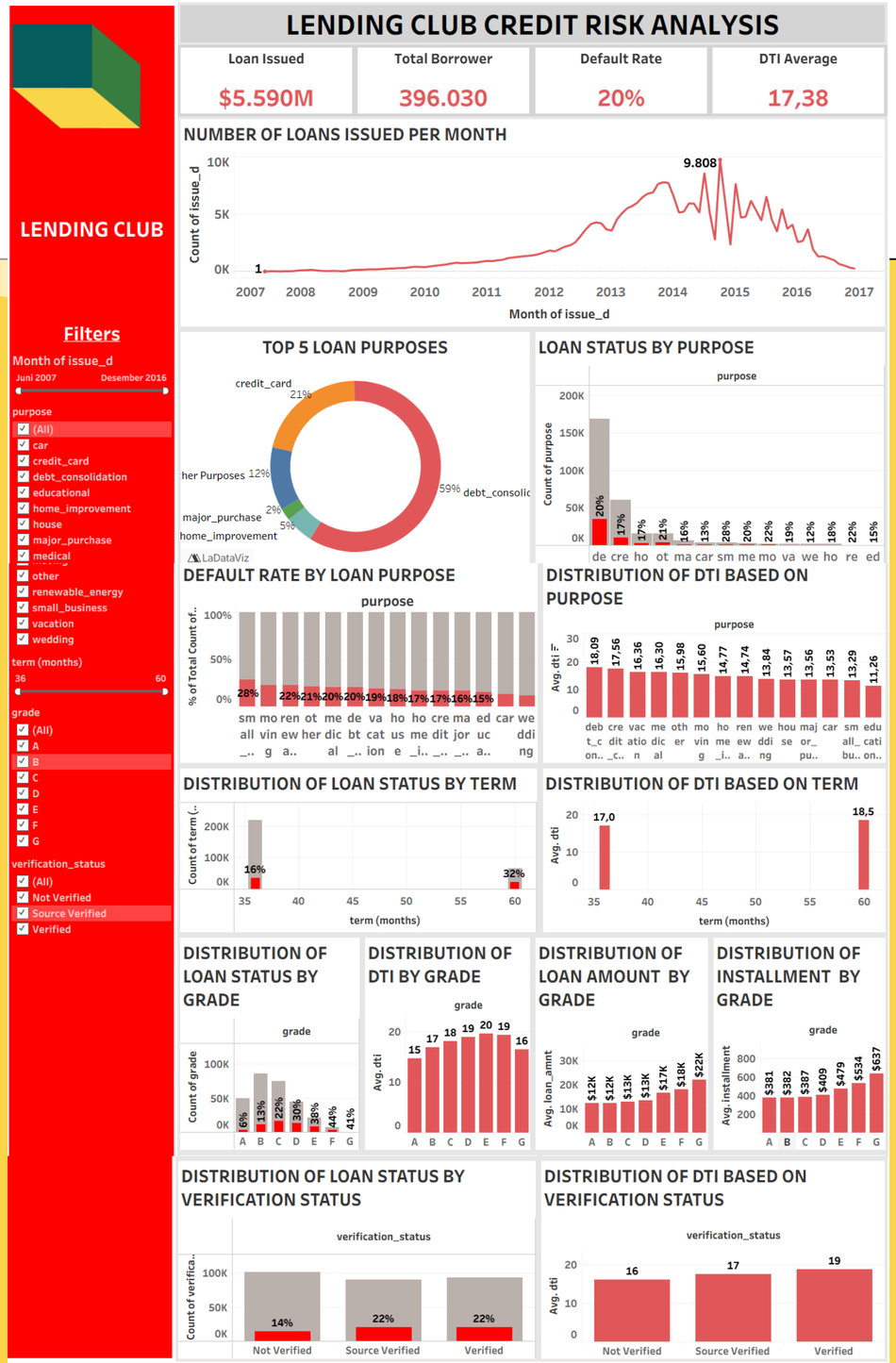
04

F1 SCORE = 73%

Balance on precision & recall

Model has good accuracy & recall, while need improvement in precision.

INTERACTIVE DASHBOARD



https://public.tableau.com/views/DEEPPW14-Milestone4AukySusanto-Salin/DASHBOARD?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

SUMMARY & RECOMMENDATIONS

Insight	Business Alignment	Recommendation
Top lending purpose is debt consolidation (> 50%) & credit card (>20%) with high default rate.	The company may consider limit loan with this purpose & exploring other lower risk purpose	1. Wedding, Car, Educational loan purposes show lower default rate with low DTI average 2. Develop Loan Product tailored to Wedding, Car, Educational purpose in collaboration with Wedding Organizer, Car Showroom, Online Course
Default rate is higher in 60 months term than 36 months	Longer term means borrower at more risk having finansial change (e.g. Losing job) during term	1. Consider discontinue 60 months term loan
Higher grade (A,B,C) show better default rate than lower grade (D,E,F,G).	Higher grade (A,B,C) already show more loan approved than lower grade (D,E,F,G) with highest loan issued for grade B & C	1. Consider reduce lower grade (D,E,F,G) loan even more

SUMMARY & RECOMMENDATIONS

Insight	Business Alignment	Recommendation
Default rate & DTI Average in not verified is lower than verified. However, overall default rate are high (>5%) in all verification status.	Verification status is a proxy for initial risk. High grade customers are often fast-tracked (Not Verified), while lower grade customers undergo mandatory verification (Verified).	1. Lower the threshold for Mandatory Verification. If an applicant's "Self-Reported" income is significantly higher than the average for their profession/location, trigger a Mandatory Verification regardless of their grade.
There was significant growth in loan disbursements from 2012 - 2014, followed by a sharp decline between 2014 - 2016.	The 2012-2014 boom reflects the "Fintech Golden Age" where rapid expansion & massive investment toward fintech company was happening. The 2014-2016 drop aligns with the LendingClub Scandal (2016) and a subsequent shift toward stricter internal loan policies to address rising defaults.	1. Use this historical trend as a "Lesson Learned." Shift the business focus from Volume-Based Growth to Quality-Based Growth by implementing the new 84% accuracy Propensity Model

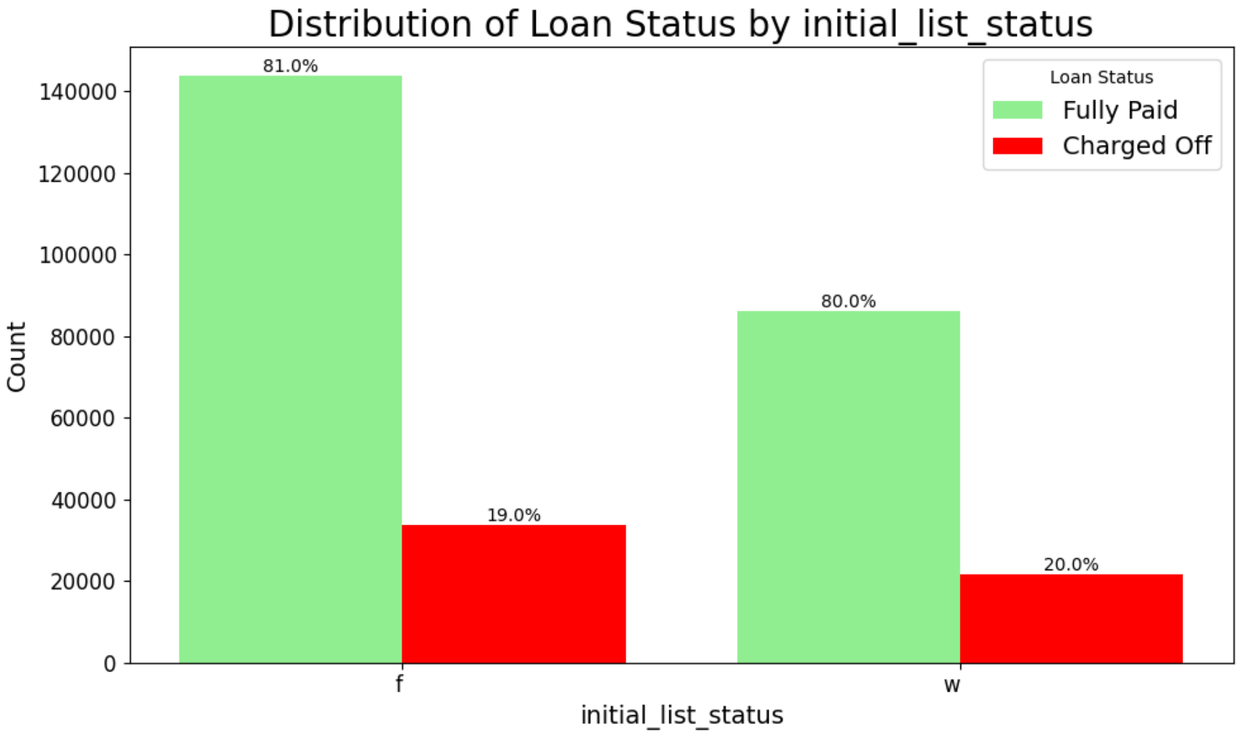
SUMMARY & RECOMMENDATIONS

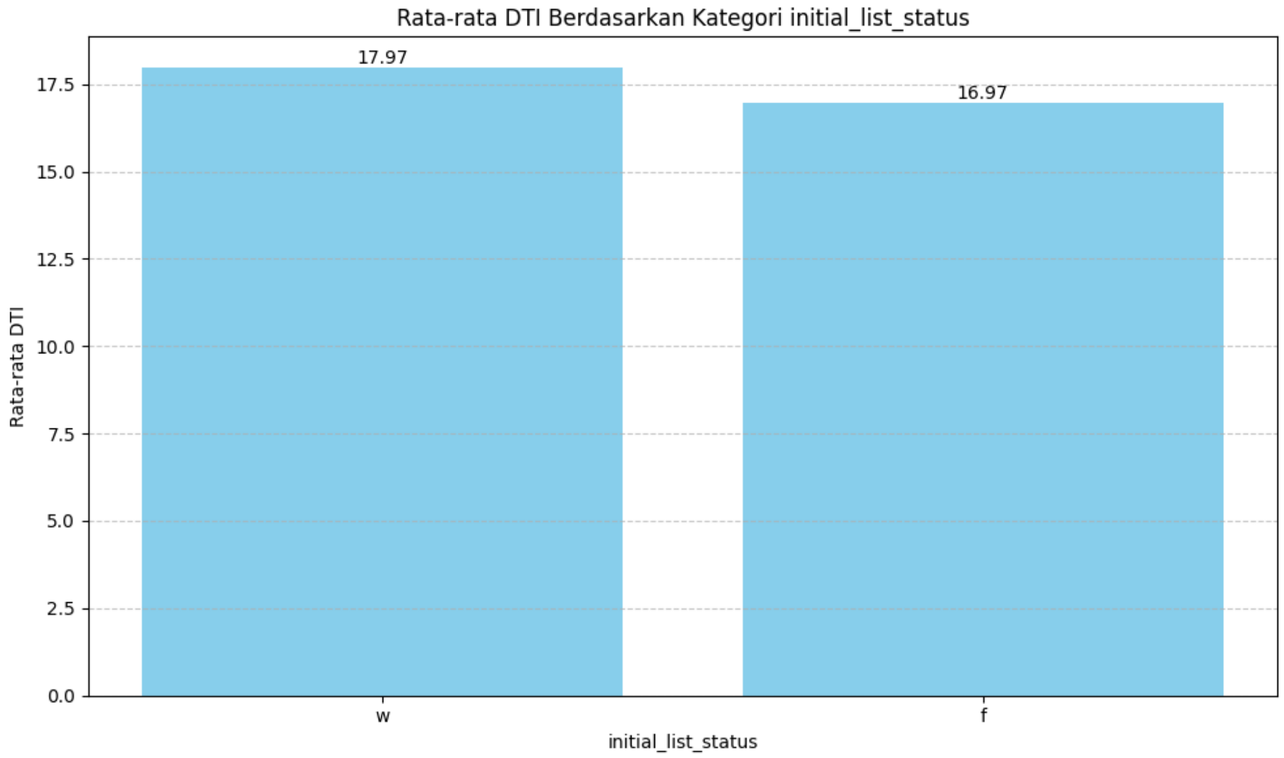
Insight	Business Alignment	Recommendation
Prospensity model using Logistic Regression method with good accuracy & recall.	Implementing Prospensity model in credit risk assesment can reduce default rate	1. Implement the model for better credit risk assesment 2. Retrain the model annually to capture new pattern

THANK YOU

END OF PRESENTATION

DISTRIBUTION OF LOAN STATUS & DTI BY INITIAL LIST STATUS

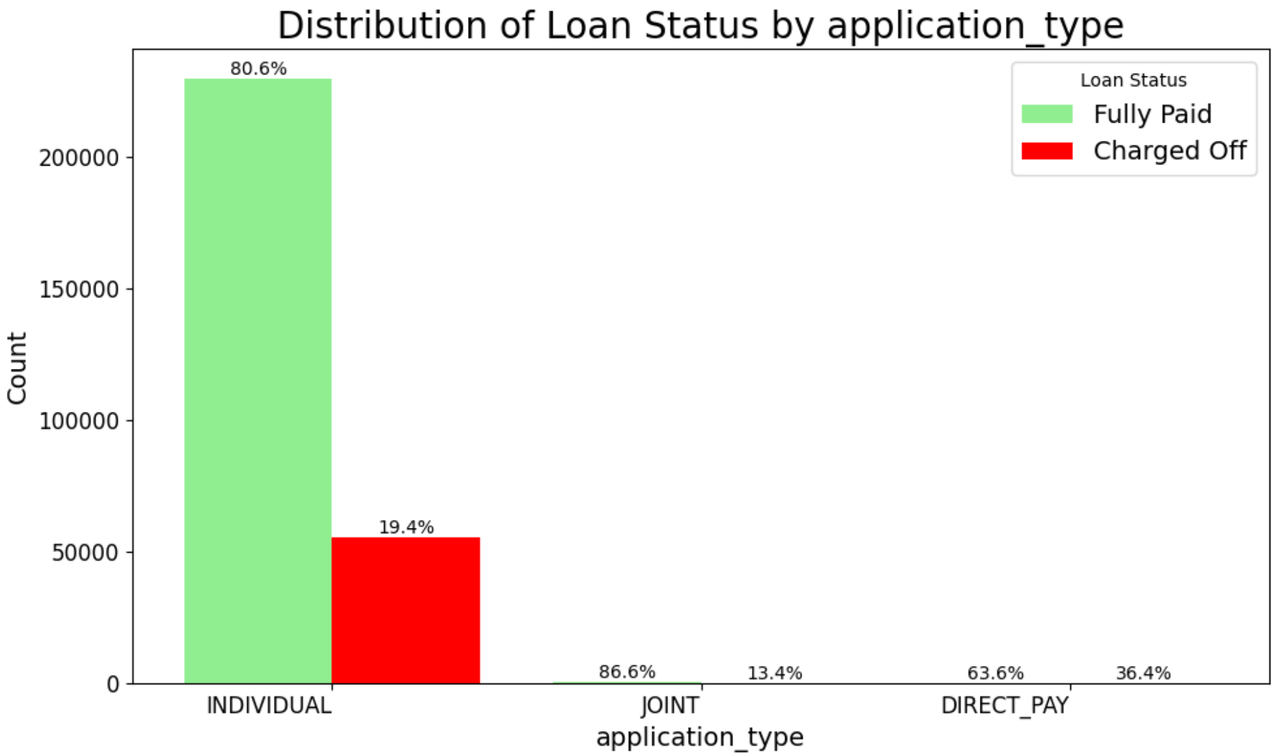


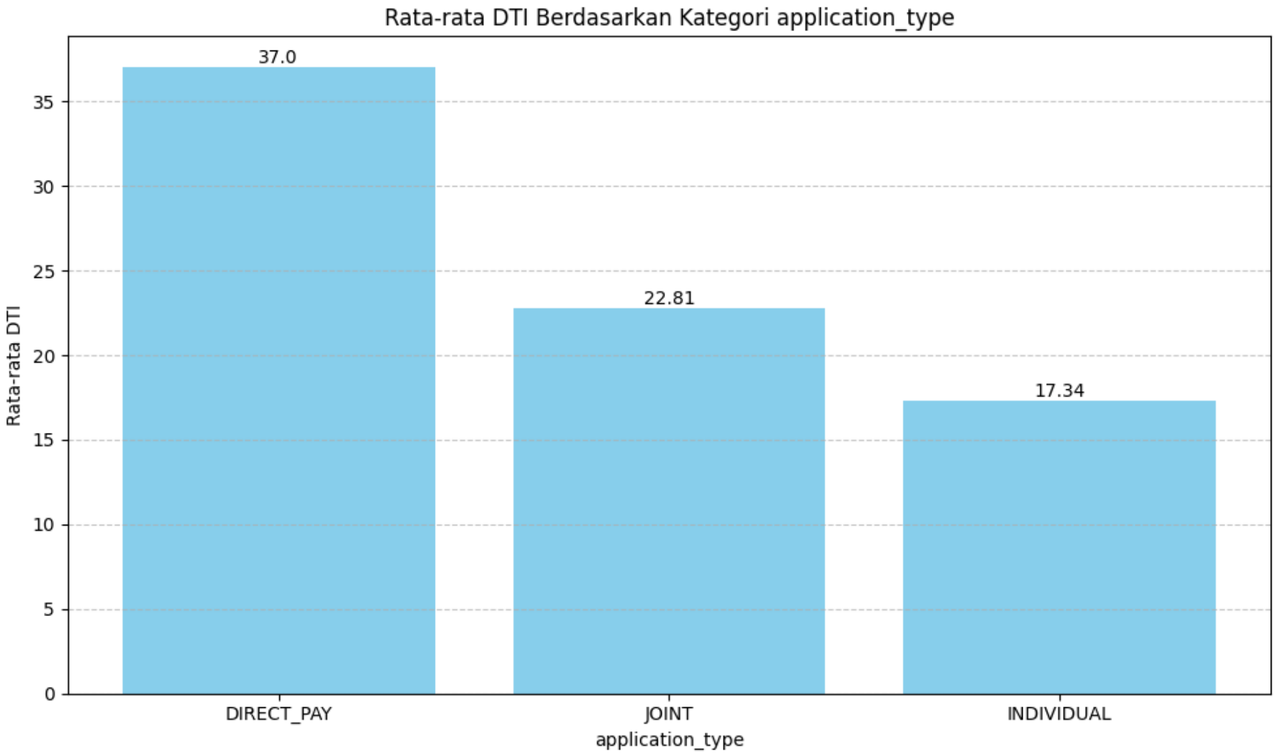


Distribution of Loan Status & Average DTI are not significantly different between initial list status.

Fractional Loan (F) is more than Whole Loan (W)

DISTRIBUTION OF LOAN STATUS BY APPLICATION TYPE

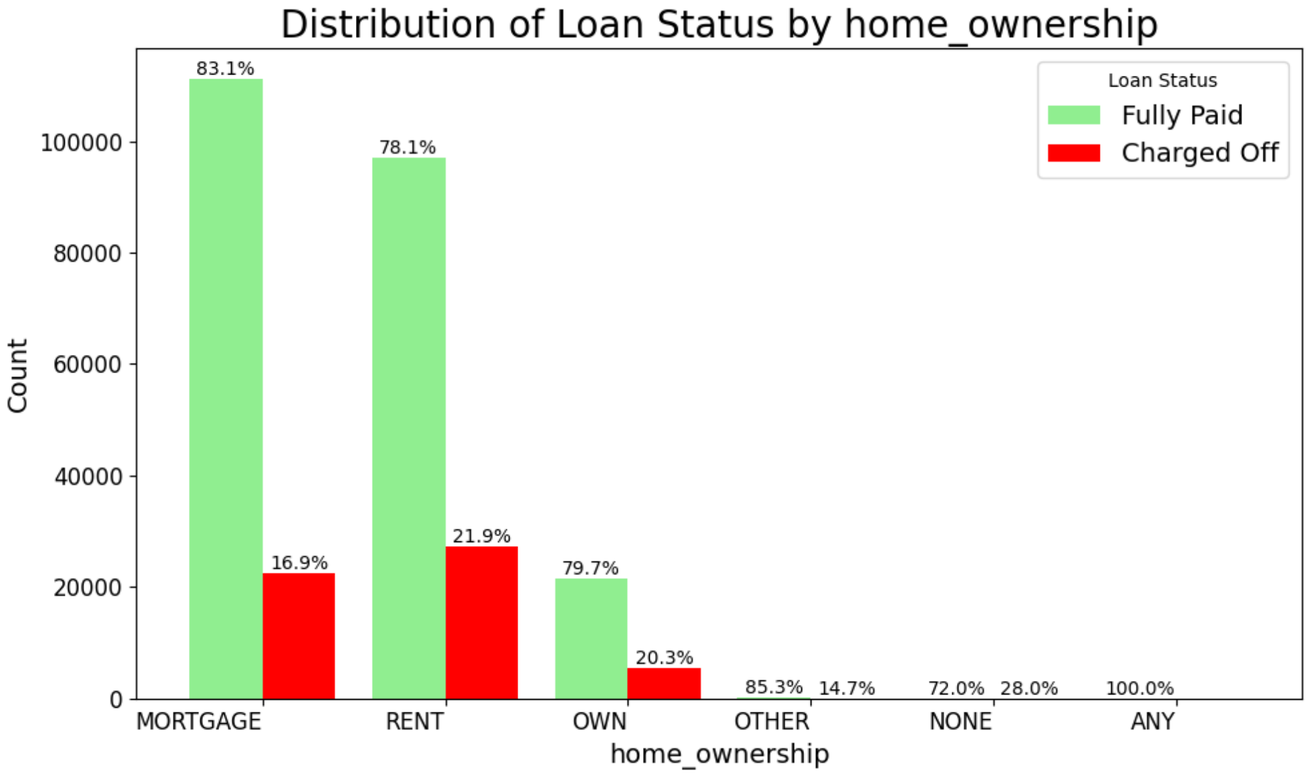


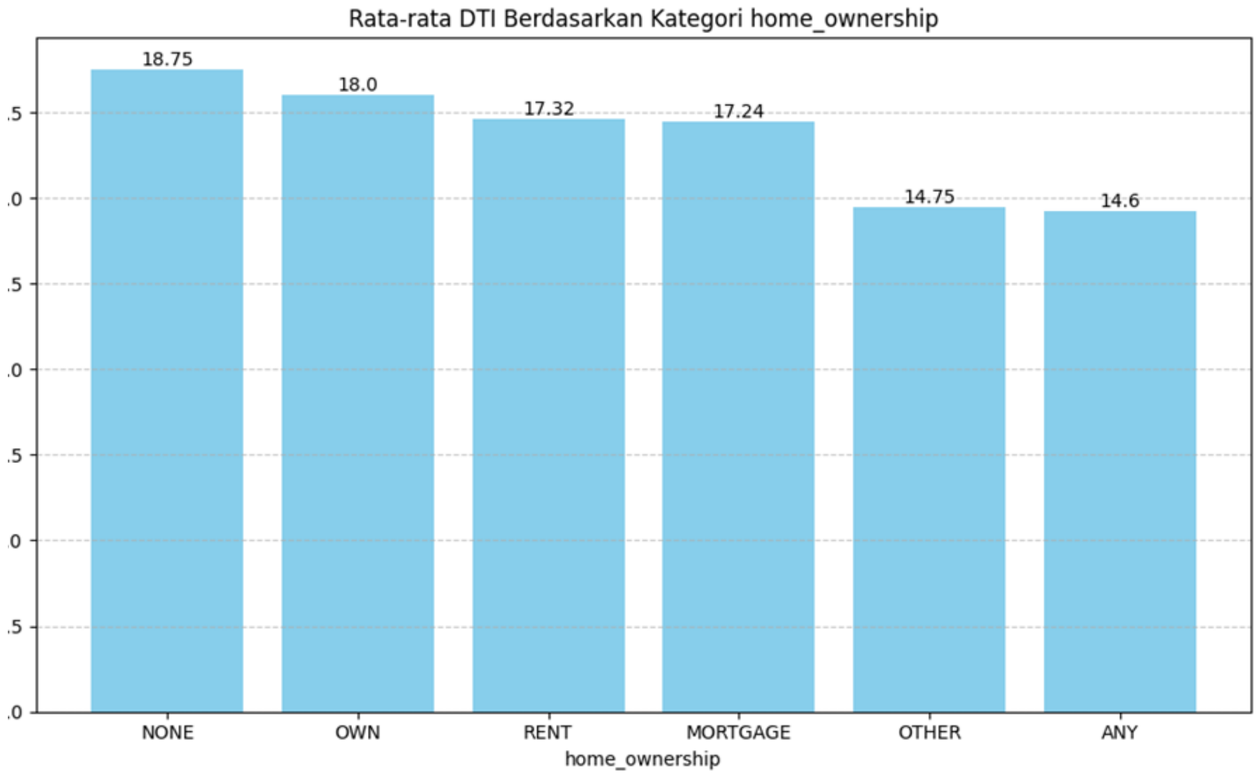


Direct Pay shows the highest charged off & DTI average, while Joint Type has better default & lowest charged off.

However, since most of customers are Individual Type, this column is irrelevant

DISTRIBUTION OF LOAN STATUS BY HOME OWNERSHIP





Loan is taken more by customers with Mortgage & Rent than customers that own house.

Default rate & Average DTI are not significantly different between customers with Mortgage, Rent, & Own